Sahan Dissanayake

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Introduction

Recent Electrical and Electronic Engineering graduate with a specialization in Signal Processing and Artificial Intelligence. Gained hands-on industry experience in enterprise data engineering, machine learning, artificial intelligence, and Radio Access Network analytics, and developed scalable, data-driven solutions for real-world engineering challenges.

Education

University of Peradeniya, BSc (Hons) in Electrical and Electronic Engineering — 2022 - 2026

- **Specialization :** Signal Processing and Artificial Intelligence
- **Relevent Coursework :** Communication Theory, Machine Intelligence and Smart Systems, Video and Image Coding, Non Linear Multi Variable Systems, Undergraduate Project.

Dharmaraja College, Kandy — 2006 – 2019

- **GCE A/L :** 3A's (2020, Island rank: 238, District rank: 18, Z-score: 2.4191)
- **GCE O/L :** 9A's (2016)

Industrial Experience

Network Analytics and Automation Intern, Dialog Axiata PLC, Colombo — April – Oct 2026

- **Data Engineering:** Designed and optimized scalable PySpark ETL pipelines in Microsoft Fabric using a Medallion architecture, and gained hands-on experience in pipeline monitoring, lifecycle management, and data lineage within a big-data environment.
- **Machine Learning:** Gained hands-on experience developing, training, and piloting an end-to-end machine learning data pipeline to detect network coverage changes.
- **Big Data Analytics:** Gained hands-on experience using SQL and PySpark to transform and integrate large-scale telecommunications data into structured analytical datasets.
- **Radio Access Network Analytics:** Gained foundational experience in Radio Access Network (RAN) analytics.

Projects

Cellular Network Coverage Changes Detection| Industrial Training — 2026

In collaboration with the Radio Network Planning unit

- Built an automated spatial data-cleaning pipeline in Microsoft Fabric for continuously updated MR/MDT telemetry, using neighbourhood-based denoising to remove isolated cells and single-cell gaps from 20 m $\times$ 20 m grids.
- Developed a data-labeling algorithm using exponential time-decay weighting across two seven-day periods to generate coverage-change labels from 14-day temporal windows.
- Trained and deployed a PyTorch ConvLSTM model in Azure ML Studio to detect spatial and temporal changes in cellular network coverage.
- Deployed an end-to-end daily pipeline that transferred cleaned data from Microsoft Fabric to Azure ML Studio, stored predictions in PostgreSQL/PostGIS, and enabled QGIS visualization of coverage improvements and degradations.
- Tools Used: Python, PySpark, SQL, PyTorch, Microsoft Fabric, Azure ML Studio, QGIS

Integrating Multi-Sensor Data and Cardiovascular System Modelling for Machine Learning Based Prediction of Cardiovascular Diseases | FYP — 2026

- Designed a non-invasive, multimodal deep learning framework for continuous cardiovascular monitoring. Extracted large-scale clinical data, processed ECG and PPG signals, and enhanced the architecture using a Physics-Informed Neural Network (PINN)-inspired loss function.
- Tools used : Python, PyTorch, Scikit-Learn

Hybrid Video Codec Design — 2026

- Developed a complete hybrid image and video compression engine that simulated modern standards such as H.264. Implemented an intelligent decision block that evaluated macroblock motion estimation (INTER) against spatial data (INTRA), and built a closed-loop optimizer that dynamically adjusted the DCT quantization scale to maximize visual quality (PSNR) for a fixed 382 kbps transmission target.

- Tools Used: Python, OpenCV

LoRA Fine-Tuning and Evaluation of a Small Language Model for Arithmetic Word Problems 2026

- Fine-tuned HuggingFaceTB/SmolLM2-360M on 6,726 GSM8K arithmetic word problems using response-only Supervised Fine-Tuning (SFT) and LoRA adapters with rank (r) 16. Developed a reproducible Kaggle GPU workflow covering data preparation, validation-based checkpoint selection, and before-and-after evaluation of numerical correctness and answer formatting.
- Tools Used: Python, PyTorch, Hugging Face Transformers, PEFT, Kaggle

Sri Lanka Trip Planner | RAG Chatbot 2026

- Developed a RAG travel chatbot that retrieved travel listings from Supabase and used Google Gemini to generate personalized recommendations and day-by-day itineraries. Implemented response validation to ensure that recommended listings matched database records.
- Integrated persistent trip preferences, PDF exports, WhatsApp sharing, and quote requests.
- Tools Used: React, TypeScript, Supabase

Multiclass CVD detection using ResNet model 2025

- Evaluated 1D Convolutional Neural Networks (CNNs) with residual architectures for automated Electrocardiogram (ECG) interpretation. Investigated how deep residual learning extracted high-level morphological features from raw waveforms to predict major cardiovascular disease (CVD) superclasses.
- Tools Used: Python, TensorFlow / Keras

Certifications

Deep Learning Specialization (DeepLearning.AI, Coursera)

- Neural Networks and Deep Learning
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
- Structuring Machine Learning Projects
- Convolutional Neural Networks
- Sequence Models

Skills

- **Programming Languages:** Python, SQL
- **Frameworks & Libraries:** PyTorch, TensorFlow/Keras, PySpark, Scikit-Learn, OpenCV, Pandas, NumPy, Hugging Face Transformers
- **Cloud & Data Platforms:** Microsoft Fabric (OneLake), Azure ML Studio
- **Data Visualization** QGIS, Power BI
- **Development Tools:** Git/GitHub, Docker, Kaggle, Google Colab
- **Soft Skills:** Problem Solving, Adaptability, Teamwork

Extra Curricular Activities

University of Peradeniya - Baseball Team 2022 - Present

- Official Team Player

University of Peradeniya - Rotaract Club

- Co-Chair of Project NextStep 360 (2023)
- Committee Member of Project EQUINOX (2022)

References

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